

Performance Testing

Date	05 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	/predict API
Test Environment	Hosted Server
Test Date	05-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a valid credit card application for prediction	10	60	All requests are processed successfully with response time below 2 seconds.
2	Submit invalid or incomplete application data	10	30	The API returns validation errors without crashing.
3	Multiple users request predictions simultaneously	50	60	The application handles concurrent requests with stable performance.
4	Peak load testing with many simultaneous requests	100	60	The application remains available with acceptable response times.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 Seconds	Pass	Within The acceptable limit
2	Response Time (Max)	< 5 seconds	3.18 Seconds	Pass	No Timeout observed
3	Throughput (Req/sec)	>50Req/sec	58 Req/Sec	Pass	Good request handling Capacity
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	46%	Pass	CPU usage Remained Stable
6	Memory Utilization	< 80%	54%	Pass	Memory usage within Acceptable range

Step 4: Observations & Analysis

Key Findings

- The application successfully handled both normal and peak user loads.
- Average response time remained below 2 seconds.
- No request failures were observed during testing.

Bottlenecks Identified

- No significant bottlenecks were identified during testing.
- The application handled concurrent requests efficiently without performance degradation.
- CPU, memory usage, and response time remained within the expected thresholds

Optimization Steps Taken

- Optimized the machine learning model to reduce prediction time.
- Implemented input validation to prevent invalid requests and improve reliability.
- Reduced unnecessary API processing to improve response time.

Step 5: Screenshots / Evidence

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 The following screenshots provide evidence of performance testing performed on the deployed Credit Card Auto Approval Prediction application using JMeter and Postman.



1. JMeter – Thread Group Configuration

2. JMeter – HTTP Request Sampler

3. JMeter – Summary Report

4. JMeter – View Results Tree (Sample Response)

5. JMeter – Aggregate Report

6. Postman – API Request (/predict)

7. Postman – API Response (200 OK)

Evidence: The above screenshots demonstrate the successful performance testing of the Credit Card Auto Approval Prediction application. The application handled 100 concurrent users with 100 iterations and achieved an average response time of 1.42 seconds, throughput of 58 requests/second and 0% error rate.